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REVIEW ARTICLES

Recommended methods for determining responsiveness and minimally important differences for patient-reported outcomes

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Abstract

Objective: The objective of this review is to summarize recommendations on methods for evaluating responsiveness and minimal important difference (MID) for patient-reported outcome (PRO) measures.

Study Design and Setting: We review, summarize, and integrate information on issues and methods for evaluating responsiveness and determining MID estimates for PRO measures. Recommendations are made on best-practice methods for evaluating responsiveness and MID.

Results: The MID for a PRO instrument is not an immutable characteristic, but may vary by population and context, and no one MID may be valid for all study applications. MID estimates should be based on multiple approaches and triangulation of methods. Anchor-based methods applying various relevant patient-rated, clinician-rated, and disease-specific variables provide primary and meaningful estimates of an instrument's MID. Results for the PRO measures from clinical trials can also provide insight into observed effects based on treatment comparisons and should be used to help determine MID. Distribution-based methods can support estimates from anchor-based approaches and can be used in situations where anchor-based estimates are unavailable.

Conclusion: We recommend that the MID is based primarily on relevant patient-based and clinical anchors, with clinical trial experience used to further inform understanding of MID. © 2008 Elsevier Inc. All rights reserved.

Keywords: Patient-reported outcomes; Health-related quality of life; Minimal important differences; Clinical significance; Anchor-based methods; Distribution-based methods

1. Introduction

Patient-reported outcomes (PROs) are frequently incorporated in clinical trials comparing health interventions for chronic diseases. These PROs include measures of health-related quality of life (HRQL), symptoms, and treatment satisfaction. PROs provide the patient's perspective and help us understand the effects of disease and treatment on symptoms, functioning, and other outcomes [1−3]. For many chronic diseases, PROs represent one of the most important health outcomes for evaluating the effectiveness of treatments and changes in disease trajectory. As far back as Hippocrates, listening to the patient has been considered an integral part of medical science [4]. Therefore, the patient's perspective of her health is integral to understanding health outcomes. The application of relevant and psychometrically sound PROs in clinical trials assists patients, their family members, and clinicians in understanding the comprehensive impact of treatment on patient symptoms, functioning, treatment preferences, and general well being.

To be useful in clinical trials evaluating new health interventions, PROs, similar to other health outcomes, must have acceptable reliability and validity [1,2,5,6]. Responsiveness is an aspect of construct validity [7] and is determined by evaluating the relationship between changes in clinical and patient-based endpoints and changes in the PRO scores over time, or based on the application of a treatment of known and demonstrated efficacy [2,5,8]. Responsiveness can be evaluated based on observational studies or in clinical trials. Evidence supporting responsiveness and

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What is new?

- Recommend that the minimal important difference (MID) be based primarily on appropriate patient-based and clinical anchors that are correlated at ⩾0.30 with the patient-reported outcome (PRO), with clinical trial experience used to further inform understanding of MID.
- MID may vary by population and context, and thus a single MID may be insufficient for all study applications involving a PRO instrument.
- Estimation of MID for a specific PRO measure should be based on multiple approaches and triangulation of methods.
- Various methods for estimating MIDs often converge, and generalizability of MID estimates for similar applications is supported.
- Recommend basing the final selection of MID values on systematic review and evaluation process such as a modified Delphi method.

for interpreting PRO results is critical for clinical trial settings. Information on the interpretation of changes or differences in PRO scores is based on the minimal important difference (MID). Demonstrating a MID is also important evidence for achieving successful PRO claims through regulatory agencies [9,10]. Nonetheless, virtually all instruments found to differentiate among clinically distinct groups are also found to be responsive to change.

Although responsiveness and interpretation of PRO measures have been discussed for the past 15 years or more [2,5,7,11−29], recommendations about the best approach for evaluating responsiveness and determining MIDs for PRO instruments are still needed. For example, the FDA requested further information and guidance on methods for determining responsiveness and MID [9]. Although there is an evolving consensus as to the best approach to evaluating responsiveness and MID [25,26,29], there is no clear statement about the recommended methods and about important issues underlying responsiveness and MID.

This report focuses on issues and recommendations for evaluating responsiveness and MID for PRO measures in chronic disease. These issues are especially germane given that for most chronic diseases cure is not feasible, and that the main objective of treatment is to maintain or improve patient functioning and well being. The remainder of this report will cover (1) conceptual issues and definitions; (2) methods for evaluating responsiveness and MID; (3) recommended decision criteria for determining MID; and (4) summary and conclusions. We will illustrate methods and concepts using published health outcomes literature.

2. Interpretation of PROs: conceptual issues and definitions

PROs require the patient to assign a response to questions (or statements) about their perceptions or activities, such as symptoms, capabilities, or performance of roles or responsibilities. These responses are typically combined in some way to create summary scores that can be used to measure concepts such as physical, psychological, or social functioning and well being, or symptom burden or severity. Symptoms can be rated based on frequency, severity, duration, degree of bother, or impact on patient activities. Demonstrating the ability to detect responsiveness to meaningful change is necessary but not sufficient for estimating the smallest change in score that can be regarded as important. This amount of change score has been referred to as the MID, and when connected to clinical anchors, sometimes as the minimal clinically important difference (MCID). Responsiveness represents the instrument's ability to detect changes whereas the MID denotes the smallest score or change in score that would likely be important from the patient's or clinician's perspective.

Because responsiveness and MID depend on population and contextual characteristics, there is not necessarily a single MID value for a PRO instrument across all applications and patient samples. There is often a range in MID estimates that varies across patient population and clinical study context.

The MID has been defined as the smallest difference in scores of a PRO measure that is perceived by patients as beneficial or harmful, and which would lead the clinician to consider a change in treatment [8,22]. A number of anchor-based and distribution-based methods have been used to determine the MID for PRO measures [22,23,25]. However, the current situation for determining the MID is fluid and evolving, and there is no clear consensus as to the recommended, best-practice approach for determining the MID [22]. Some have recommended to estimate the MID based on several anchor-based methods, with relevant clinical or patient-based indicators, and to examine various distribution-based estimates (i.e., effect size, standardized response mean, standard error of measurement [SEM]) as supportive information, and then to triangulate on a single value or small range of values for the MID [28−31]. Similar to virtually every measure used in medicine, all PRO assessments include some measurement error. The amount of error in most well-developed PRO instruments is similar to or less than what is observed in most standard clinical measures [32]. One needs confidence that observed changes in scores over time with treatment are not primarily attributable to error. Put another way, instrument reliability helps ensure that observed differences between treatment groups exceed what one might expect based upon measurement error alone [21,32,33]. Confidence in a specific MID value evolves over time and is confirmed by additional research evidence.

2.1. Minimal versus meaningful differences

The idea of a MID came about in the literature somewhat indirectly. In efforts to evaluate an asthma quality of life questionnaire, Jaeschke et al. [11,34] attempted to estimate how much of a difference in scores would result in some change in clinical management that is to be considered clinically meaningful. They ''developed an approach to elucidating the significance of changes in score in quality of life instruments by comparing them to global ratings of change. Using this approach (they) established a plausible range within which the minimal clinically important difference (MCID) falls'' [11]. It is important to note that Jaeschke did not define an MCID as much as indicating that the technique they used was a reasonable approach to produce an estimate of the interval within which an MCID falls. They concluded that such an approach provided support for an estimate for a clinically meaningful difference that could be applied to both groups and individual patients. They, however, did not claim to actually derive a truly ''minimal'' difference estimate.

More recently, Sloan proposed a distribution-based method (the empirical rule effect size or ERES method) from a perspective of finding an effect size that was nonignorable [35,36]. Sloan noted that many of the competing methods seemed to converge into the same general area in terms of proportion of standard deviation (SD) of the PRO instrument under study [37,38]. The MCID derived by Jaeschke et al. [11,34], for example, was precisely equal to the ½SD estimate produced via the ERES method. Further data by Norman et al. [24] seemed to confirm this hypothesis that ½SD could indeed be a minimally important difference in some circumstances. Sloan was not searching for a *minimum* but was instead looking for a conservative estimate for a clinically meaningful difference (i.e., an ''obviously'' important difference). Recent work has indicated that there are situations where a difference smaller than ½SD may be meaningful [27,36]. Evidence supporting any MID is needed to justify such estimates, and sensitivity analysis using multiple approaches is recommended wherever possible. Where no such evidence is available, the ½SD estimate may be a reasonable place to start as a meaningful difference.

The idea of finding an estimate of the ''minimum'' important difference is at once intuitively appealing and mathematically challenging. Some have proposed alternative terminology (summarized in Sloan et al. [36]), but the nomenclature is still inconsistent. Can we ever get to a true minimum? It is more reasonable to assume that we can get to a consensus on what is meaningful for practical purposes and use this as a benchmark. Determining when a minimum has been achieved is harder to gauge and requires greater precision. Thus, the MID may only be useful to sort cases as improved or not (or as worsened or not), with the substantial benefit judged based on a sizable increase in the proportion of improved cases (or sizable decrease in the number of declined cases) observed by treatment group.

3. Methods of evaluating responsiveness and clinical significance

Longitudinal studies are needed to determine whether a PRO instrument is responsive to changes or differences. These studies may be randomized clinical trials comparing treatments of known efficacy or observational studies where patients are treated with usual medical care and followed over relevant periods of time. For clinical trial designs, there needs to be some evidence that the treatment is effective and that the expected changes in clinical status are linked to expected changes in the PRO measure. Careful attention to selecting treatments for clinical trials will attenuate circumstances where the researcher may falsely conclude that the PRO is not response, because the PRO and clinical endpoints are not associated with each other. To assess responsiveness, some criterion is needed to identify whether patients have changed (either improved or worsened) over time. These criteria, or anchors, may be clinical endpoints, patient-rated global improvement, change in other PRO measures, or some combination of clinical and patient-based outcomes. In addition to the anchor-based approaches, responsiveness and clinical significance can be informed by previous work using distribution-based methods and through systematic reviews of clinical trials.

3.1. Anchor-based methods

The anchor-based approaches use an external indicator, either clinical or patient-based, to assign subjects into several groupings reflecting no change, small positive changes, large positive changes, small negative changes, or large negative changes in clinical or health status. The anchors can be clinical (i.e., laboratory measures, physiological measures, and clinician ratings) or patient based, such as global ratings of change or actual changes in PRO measures that have demonstrated MID in the target patient population. It is strongly recommended to use multiple independent anchors and to examine and confirm responsiveness across multiple samples [22,25,28,29,39,40].

Selecting anchors should be based on criteria of relevance for the disease indication, clinical acceptance, and validity and evidence that the anchors have some relationship with the PRO measure. The best anchors for estimating the MID—whether retrospective measure of change, knowledge about the course of health over time, or clinical parameters—are ones that identify those who have changed to a small but meaningful degree. Including individuals who change beyond a small but meaningful degree risks over-estimating the MID. It is important to identify the subset of people who have experienced minimal change. Those patients who have changed by a minimal amount have been

identified by asking study participants at follow-up to report how much they changed since baseline of a study using a multiple categorical response scale. People who reported either getting *a little better* or *a little worse* constitute the minimal change subgroup. The change in PRO measures reported by this subgroup is the estimate of the MID as perceived by the patient. One can decide to examine change for those getting worse versus getting better separately or pool them together after accounting for the difference in the direction of change.

Retrospective self-reports (such as health transition questions) are known to be subject to recall bias [41]. When retrospective change items are used as anchors, it is useful to determine if they reflect the baseline (pretest) and present (posttest) status equally. In theory, retrospective change items should correlate positively with the posttest and have a negative correlation of equal magnitude with the pretest as illustrated in the following formulas: $r(x, y - x) = r(x,y)$ and $r(y, y - x) = r(y,-x) = -r(x,y)$, where $r(.,.)$ is the correlation, x is the pretest, and y is the posttest. In reality, retrospective self-reports tend to correlate more strongly with the posttest than they do with the pretest because current status unduly influences the retrospective perception of change. For example, Walters and Brazier [42] found moderate correlations (mean 0.45, range: 0.18−0.57) between responses to a retrospective measure of global change and the SF-6D at follow-up across nine studies. Correlations with initial assessments were systematically lower (mean 0.22, range: 0.01−0.41). Thus, these correlations should be interpreted with flexibility and allowance for lack of equality.

For a clinical parameter, it is also necessary to establish the amount of change on the anchor that is a reasonable indicator of minimal change. Estimating the MID requires agreement about what constitutes a minimal change in the anchor. Kosinski et al. [39] defined minimal improvement on their clinical measures as 1%−20% improvement in the number of swollen and tender joints in a study of patients with rheumatoid arthritis. Although this may be a reasonable threshold, other investigators might argue for another threshold (e.g., >10%; >20% improvement). Any anchor that is chosen should have a ''nontrivial'' association with change in the PRO measure. If the correlation between the anchor and PRO change is zero, then the anchor is not useful for establishing the MID. While a nontrivial correlation is important, a clinical anchor cannot ''hope to capture the richness and variation of the construct of HRQL'' [17]. Using Cohen's [43] rules of thumb, we recommend 0.30−0.35 as a correlation threshold to define an acceptable association between an anchor and a PRO change score, although alternative thresholds may be acceptable in the presence of supplementary information.

The variety of possible anchors and uncertainty in the anchor cut-point that defines a minimal difference make a single estimate of MID problematic. Using the retrospective report anchor as an example, the recall item might refer globally to change in ''health,'' ''health-related quality of life,'' or ''quality of life.'' Moreover, the anchor might be worded more specifically such as ''physical functioning,'' ''pain,'' etc. The choice of words can lead to variability in the performance of the anchor. Any specific anchor may be more or less appropriate for different PRO domains. For example, an energy/fatigue scale might be expected to change more than a pain scale in response to change in hematocrit [44].

There also needs to be an understanding of the trajectory of health outcomes in the target disease to evaluate responsiveness. For example, do most patients improve over time with treatment, as with seasonal allergic rhinitis, or as in many chronic diseases (e.g., COPD, arthritis, etc.), is the expected trajectory one of maintenance or varying levels of deterioration in health status over time, even with treatment? Other factors that can lead to variation in the estimation of the MID include whether the people being evaluated are high or low on the measure at baseline, whether they improve or decline in HRQL over time, and whether they have similar demographic, clinical, and other characteristics [18].

Once groups of patients are identified as improving, worsening, or remaining stable based on several relevant external anchors, several data analyses and indicators can be used to examine responsiveness. First, analysis of variance or covariance procedures can be performed comparing differences in mean baseline to endpoint changes in the PRO scores across the meaningful change groups (i.e., stable versus small improvement, stable versus moderate improvement, etc.). Second, responsiveness to change is frequently evaluated using different indicators [5,33], such as the effect size (ES) [45], standard response mean (SRM) [46], and responsiveness statistic (RS) [8].

3.2. Previous clinical trial experience

As the medical literature on PROs applied in clinical trials increases [47], it is increasingly possible to understand responsiveness and MID for different PRO instruments based on demonstrated differences between active and placebo treatments or between two or more active treatments in clinical trials. Systematic reviews of the clinical trial literature can therefore be used to determine clinical significance. For example, Niebauer et al. [48] in a systematic review found consistent evidence that omalizumab resulted in a 0.30 point improvement (i.e., 0.30 effect size) in asthma quality of life questionnaire scores when used as add-on therapy in moderate to severe asthma. There are observations that the MID determined through anchor-based methods seen in observational, psychometric evaluation studies may differ from MIDs seen in randomized clinical trials (D. Patrick, personal communication, February 23, 2006). As evidence accumulates in clinical trials, the observed changes or difference in PRO measures based on effective treatments provides a rich and valuable source of data on responsiveness and clinical significance. For example, Jones [49] summarized the clinical trial experience for

the St. Georges Respiratory Questionnaire (SGRQ) in the chronic obstructive pulmonary disease literature. He found that across eight different studies, the recommended MID for the SGRQ total score was approximately 4 points (i.e., 0.32 effect size).

Therefore, it is recommended that the previous clinical trial literature for the targeted indication be reviewed and synthesized to identify evidence on responsiveness and MID for the selected PRO instruments. These PRO data provide an excellent resource for understanding the application and performance of PRO measures in the clinical trial environment and can be used to further support the evidence base on responsiveness and MID for interpreting PRO data.

3.3. Distribution-based methods

Distribution-based methods convey a notion that a MID can be estimated based on the distribution of observed scores in a relevant sample. Guyatt et al. [22] provide concise and complete exposition of the various distributional methods. Some have criticized distribution-based methods because they are anchor free (i.e., ''meaning-free''). Still others have expressed concern over accepting a ''purely'' statistical argument to support the choice of a MID, which is interesting when statistical theory underlies virtually all clinical investigation [50].

Some researchers have suggested that the ½SD estimate [24] or that the SEM [51,52] may approximate a MID for some PRO instruments. Although this magnitude of change is certainly clinically significant and meaningful, it is not necessarily minimal. Empirical evidence from previous studies, physiological arguments, and statistical theory shows a tendency to converge to the ½SD criteria as being meaningful to patients [40]. Although different distribution-based indicators demonstrate that change has occurred and provide some insight as to whether the change (responsiveness) is small or large, the indices do not necessarily inform as to whether the observed change is MID. To determine MID, it is necessary to get information as to whether the observed change is important from the patient's or clinician's perspective [40]. MIDs have been observed, however, to be as small as 0.25 to 0.33 ES (or SD units) [27,29].

The distribution-based indices provide no *direct* information about the MID. They are simply a way of expressing the observed change in a standardized metric. This makes it possible to compare change observed for measures that have a different raw metric and the degree of deviation (individual and group level) within the sample. ES estimates can be compared to Cohen's guidelines about the magnitude, but anchor-based methods are the only way to estimate the MID directly. The SEM has been proposed as a method of relevance to MID estimation. This suggestion is based on anecdotal observations that the SEM was approximately equal to the estimated MID [51]. Norman et al. [24] note that 1 SEM is approximately the same as a 0.5 difference on a seven-point scale and 1 SEM is approximately ½SD when the reliability is 0.75. But why should 1 SEM have anything to do with the MID? The SEM measurement is estimated by the product of the SD and the square root of 1-reliability of a measure. The SEM is used to set the confidence interval (CI) around an individual score, that is, the observed score plus or minus 1.96 SEMs constitutes the 95% CI. In fact, the reliable change index proposed earlier by Jacobson and Truax [12] is based on defining change using the statistical convention of exceeding 2 standard errors.

Distribution-based methods are most applicable when the goal of estimating a clinically meaningful difference does not have a heavy reliance on the estimate needing to be minimal. There is a concern about the use of a conservative estimate for MID as it may require setting a criteria for success beyond what is achievable for a given treatment. Therefore, we recommend that the distribution-based measures are used as supportive information for MID estimates from different anchor-based approaches and systematic reviews of the clinical trial literature.

3.4. Determining the MID for PRO instruments

For interpreting differences or changes in PRO instruments, information needs to be provided as to whether the changes seen in the scores are important from either the patient's or clinician's perspective. The clinical meaningfulness of the observed change is based on that change perceived as minimally important and as beneficial or harmful from the patient's viewpoint. It is recommended that the patient's perspective be given the most weight, because these are PROs, although the clinician's perspective is considered important as well. The MID is best estimated using multiple anchors with the same external criteria used to evaluate responsiveness of the PRO measure. However, there are differences in how these data are used and compared to determine MID. Because the focus is on determining the MID, it is necessary to identify the smallest difference or change that is important to the patient.

In many cases, global assessments of change in health or clinical status are used to categorize patients into groups that reflect, based on their own reports, different amounts of change in the construct of interest. Most often the MID is determined as the change observed in the small improvement group, as long as the small improvement group demonstrates changes that are larger than the stable group. If the changes are similar, there is uncertainty about the group definition, change in the PRO measures, and the MID. There are cases where there is some variation observed even among the stable group, in these cases it is informative to examine the difference in mean baseline to endpoint change scores between the stable group and the small improvement (or worsening) group.

Note that there is evidence that there is asymmetry in worsening and improvement in PROs depending on the

specific disease [20,27]. Clinician global assessments of change in clinical status or evaluations of clinical severity, clinical response criteria (i.e., American College of Rheumatology [ACR] response criteria), or other indicators can be used to determine MID. For these clinical anchors, it will be necessary to identify, based on previous research or clinical consensus, what a small and clinically meaningful effect may be based on these measures. For example, in rheumatoid arthritis, the differences between groups of stable patients and those experiencing a 20% ACR response can be used to determine the MID of a PRO score. If multiple anchors are used, there will be several different estimates of MID derived corresponding to these different anchors, and the result will be a range of MID estimates for the targeted PRO instrument.

4. Recommended decision criteria for determining MID

The application of multiple methods to determine the MID for a PRO instrument in a specific patient population will almost always result in a *range* of values for the MID. This is the essence of triangulation, that is, examining multiple values from different approaches and hopefully converging on a small range of values (or one single value). It is recommended that the different MID estimates be graphed to visually depict the range of estimates. Figure 1 provides a summary of MID estimates from a study by Yost et al. [27]. To identify a single MID value (or narrow range of MID values), it is recommended that the anchor-based estimates be assigned the most weight, and experience from clinical trials be used to further support and perhaps further narrow the range of values. Interpretation of the MID from different anchors should also take into account the proximity of the anchor to the target PRO measure, that is, assign more importance to MIDs generated from more closely linked concepts.

A systematic consensus process involving several clinicians and health outcome researchers is recommended and can be completed, based on Delphi methods, to arrive at a single MID value, or at least a narrower range of values. There is no consensus as to how much data are needed as supportive evidence for the MID of a PRO instrument. Clearly, the more data and evidence the better, but a single, generalizable study with multiple patient-based and clinical anchors may be sufficient. As with other aspects of construct validity, responsiveness and the MID value are confirmed based on accumulating evidence from multiple studies and, with additional data, we can be more confident in the MID value. It would be rather unusual for a single MID to be appropriate for all applications and across all patient populations. For example, the MID derived for an asthma-specific quality of life measure in mild to moderate asthma patients may not be generalizable to clinical trials comparing an add-on treatment for patients with moderate to severe asthma [48].

5. Summary and conclusions

For PRO endpoint data to be accepted as evidence of treatment efficacy there must be evidence documenting the instrument's conceptual framework, content validity, and psychometric qualities. For responsiveness, it is necessary to demonstrate that the PRO scores are sensitive to actual changes in health status. Although demonstrating

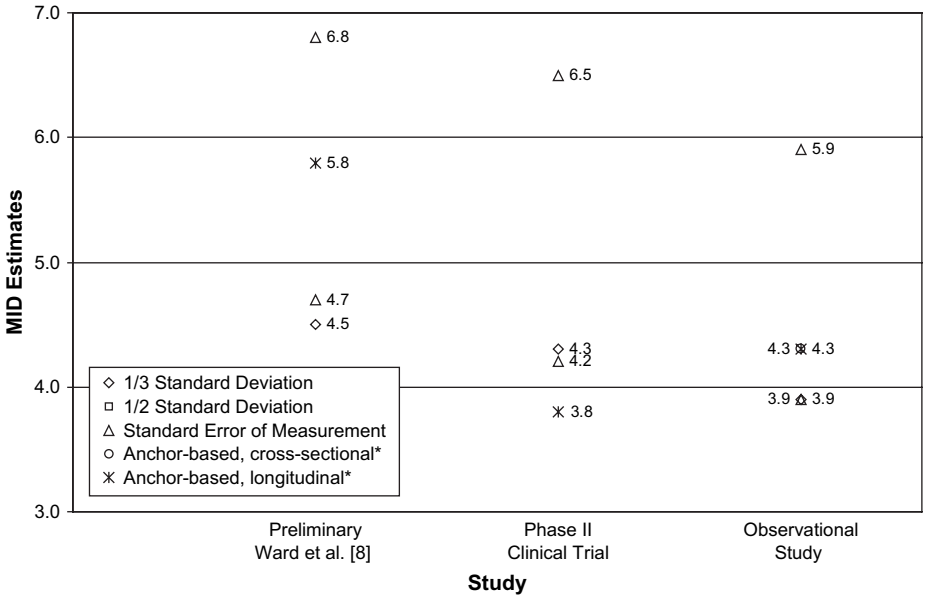


Fig. 1. Summary of Distribution- and Anchor-Based Estimates of Minimally Important Differences (MIDs) for the Trial Outcome Index-Colorectal (TOI-C). Only anchor-based values with effect sizes between 0.2 and 0.5 were plotted. *If more than one value was available for a given type of estimate, the average was plotted.

responsiveness is a key component to establishing an instrument's construct validity, it is also important to determine the MID to assist in interpreting statistically significant PRO results in clinical trials. In addition, the MID for a PRO instrument that is specified as a primary or important secondary endpoint is clearly useful for calculating statistical power and for determining sample sizes for clinical trials. The MID may vary by population and context, and no one MID may be valid for all study applications involving a PRO instrument. Responsiveness and MID must be demonstrated and documented for the particular study population.

The estimation of MID for a specific PRO measure should be based on multiple approaches [22,40] and triangulation of methods. Anchor-based methods applying various relevant patient-rated, clinician-rated, and disease-specific variables provide primary and meaningful estimates of an instrument's MID. Previous results, including the PRO measures, from clinical trials can also provide insight into observed effects based on treatment comparisons, and these data should be used to help determine MID. Distribution-based methods can support and help interpret estimates from anchor-based approaches and can be used in situations where anchor-based estimates are unavailable. We recommend that the MID be based primarily on relevant patient-based and clinical anchors, with clinical trial experience used to further inform understanding of MID.

Multiple approaches to estimating the MID will produce a range of different values, and decision guidance is needed to select a single value or narrow range of MID values. Based on examination of the resultant MID estimates, it is often possible to select a narrow range of MID estimates. When this is difficult or there is some uncertainty about which MID may be best, we recommend basing the final selection of MID values on some systematic review and evaluation process such as a modified Delphi method.

The MID for a PRO instrument is not an immutable characteristic; it may vary across populations and treatments. Incremental changes in the PRO measure may be influenced by disease severity and the treatment context. Continuing experience with the PRO instrument in different population contexts and clinical trial situations will inform the understanding of MID for the instrument. As with construct validity, accumulating evidence across multiple studies will help clinicians, and health outcomes researchers gain confidence in interpreting important differences for the PRO measure.

Selecting a MID estimate for clinical trial planning or for interpretation of PRO endpoint findings should be based on the existing knowledge on MID for the specific PRO instrument. There needs to be a clear rationale for the selection of the MID value especially in situations where there may be range of MID estimates for the instrument. It is best to select the value based on clinically relevant anchors that are proximal to the concept(s) measured by the PRO instrument, and based on the understanding of the disease area and patient population. For interpreting PRO results from a clinical trial, the proposed MID value should be identified a priori in the PRO statistical analysis plan. The rationale for selecting the MID value for a study should be clear and straightforward based on the existing evidence. If there is uncertainty about this value, additional supportive research may be needed to strengthen the evidence base.

PRO measures provide the patient's perspective on the impact of disease and treatment. These health outcomes allow for a more comprehensive evaluation of a medical intervention and require evidence on sound psychometric characteristics, including responsiveness and interpretation guidelines. MID estimates, if based on systematic research and relevant anchors, provide the basis for interpreting clinical trial results and help regulatory agencies, clinicians, and patients understand the effects of treatment of symptoms, and patient functioning and well being.

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